Hydroxyapatite XRD Data Analysis

Experimental Diffraction Characterization, Rietveld Refinement, and Theoretical VESTA Simulation

EP24BTECH11024-EP24BTECH11033

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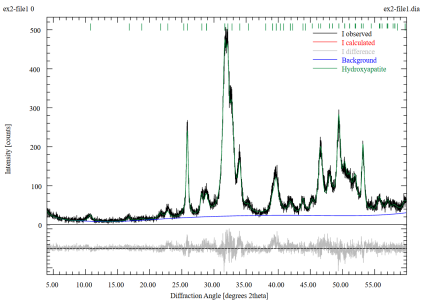


Figure 1: PROFEX XRD pattern.

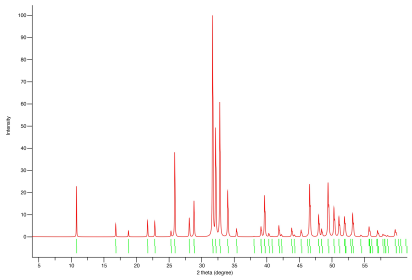


Figure 2: VESTA XRD pattern.

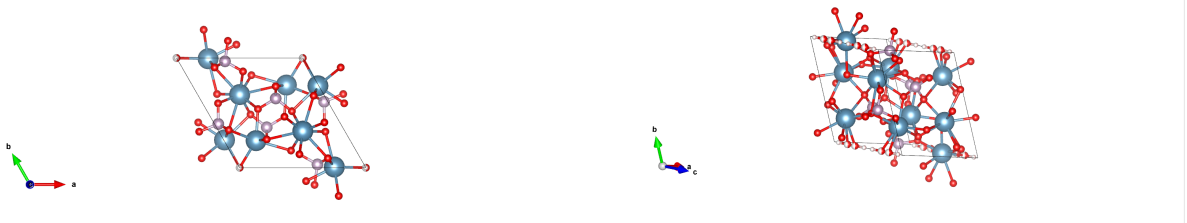


Figure 3: Hydroxyapatite VESTA 3D model shown with different orientations.

Table 1: Important parameters including unit cell lattice parameters and refinement details.

Parameter	**Value**	**Parameter**	**Value**
Formula	$Ca_5(PO_4)_3(OH)$	Space Group	$P6_3/m$ (No. 176)
Crystal System	Hexagonal	Formula Units (Z)	2
$a = b$ Lattice Parameter	9.45600 Å	$\alpha = \beta, \gamma$	$90°, 120°$
c Lattice Parameter	6.88477 Å	Unit Cell Volume (V)	533.13 Å^3
Crystallite Size $[001]$	26.14 nm	Refinement Fit (R_{wp})	13.72%
Crystallite Size $[100]$	10.48 nm	Goodness of Fit (χ^2)	1.2362

Abstract

This report presents a comprehensive X-ray diffraction (XRD) structural analysis of crystalline hydroxyapatite ($Ca_5(PO_4)_3(OH)$). The investigation encompasses raw dataset parsing from PANalytical XRDML files, experimental profile modeling using BGMN/Profex Rietveld refinement, unit-cell geometry calculation, dynamic instrument geometry corrections (such as automatic divergence slit effects), anisotropic crystallite size estimation via Scherrer broadening, and comparative validation against theoretical patterns simulated in VESTA.

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1 Hydroxyapatite XRD Data Analysis (`ex2-file1.xrdml`)

This section outlines the basic setup, data architecture, experimental parameters, and analytical workflow for processing X-ray diffraction measurements recorded for a synthesized Hydroxyapatite powder sample.

1.1 What is an XRDML File?

An **XRDML** file is an industry-standard data format created by PANalytical for storing X-ray diffraction measurements and associated metadata.

- **Structure:** The format is based on XML (eXtensible Markup Language). It organizes information hierarchically using standard text tags. This structure provides high readability for both automated analytical software and human researchers while ensuring complete separation between measured raw intensities and instrument configuration metadata.

1.2 Important Data Fields and Metadata

Opening an XRDML file reveals two main categories of information:

- **Metadata (Experimental Setup):** Contains background parameters describing the instrument and run setup. Key parameters include sample identifier, X-ray generator settings (voltage and current), source radiation wavelength (typically Copper Kα), and scan limits (start angle, end angle, and step size).
- **Data Fields (Experimental Results):** Contains two core numerical arrays: the diffraction positions (2θ angles) and the recorded intensities (photon counts measured at each step).

1.3 Description of the Experimental XRD Data

- **Collection Process:** Inside the diffractometer, a focused X-ray beam strikes the hydroxyapatite powder sample. As the goniometer rotates, the beam angle and detector position change synchronously. Crystal lattice planes scatter the X-rays, yielding constructive interference peaks at specific angles matching Bragg's Law.
- **Diffractogram Representation:** Data are plotted as intensity versus diffraction angle 2θ. Peak positions and relative heights form a unique structural fingerprint of the sample.

1.3.1 Experimental Dataset and Measurement Conditions

The experimental dataset analyzed here is recorded in `ex2-file1.xrdml`. Measurements were performed using a copper (Cu) anode X-ray tube operated at 45 kV and 40 mA. The characteristic radiation wavelengths are $\lambda(\text{Cu K}\alpha_1) = 1.540598$ Å and $\lambda(\text{Cu K}\alpha_2) = 1.544426$ Å.

The scan covers a 2θ angular range from 4.0075° to 59.9875° with a step increment of 0.015° and a counting time of 1.0 second per point, yielding 3,733 discrete measurement points.

1.3.2 Description of the Experimental Diffraction Pattern

The raw diffraction pattern was imported into Profex and plotted as intensity versus 2θ. The pattern displays distinct crystalline reflections above a continuous background baseline. A prominent reflection occurs near 25.9°, while the strongest reflections cluster between 31° and 34°. Additional reflections appear at higher angles around 39°–40°, 46°–50°, and 53°, confirming a crystalline hydroxyapatite structure.

Peak positions relate to lattice interplanar spacings (d-spacings) through Bragg's Law:

$$n\lambda = 2d \sin\theta \tag{1}$$

where n is diffraction order, λ is wavelength, d is interplanar spacing, and θ is the Bragg angle.

The measured pattern acts as the reference baseline for Rietveld profile fitting. Refined parameters are exported as a Crystallographic Information File (CIF) for 3D visualization in VESTA and comparison against theoretical models.

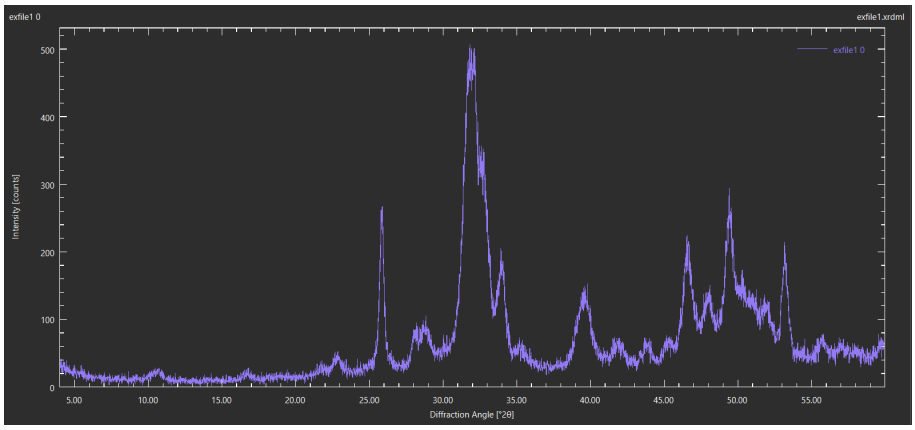


Figure 4: Experimental XRD pattern from `ex2-file1.xrdml`, plotted as intensity versus 2θ.

1.4 Common Applications of XRD

X-ray diffraction serves as a primary non-destructive characterization method:

- **Phase Identification:** Matching measured 2θ reflection profiles against reference databases (ICSD, PDF) to confirm single-phase purity.
- **Impurification & Secondary Phase Detection:** Identifying trace crystalline secondary phases (e.g., tricalcium phosphate or calcium oxide).
- **Crystallite Size Analysis:** Estimating average domain sizes from peak broadening profiles.

1.5 How to Import Data into Profex

Profex natively parses PANalytical XRDML files:

1. Open the Profex application.
2. Select **File** $\rightarrow$ **Open Raw Data File...** (or click the toolbar folder icon).
3. Select `ex2-file1.xrdml`.
4. Profex automatically reads XML metadata tags and displays the diffractogram, ready for Rietveld refinement.

1.6 Additional Context: Why Hydroxyapatite?

Hydroxyapatite ($Ca_5(PO_4)_3(OH)$) forms the inorganic mineral phase of human bones and dental enamel. XRD characterization is central to biomaterial synthesis, ensuring lab-made synthetic bone grafts and implant coatings match natural biological apatite structure.

2 Analysis of the Refined XRD Pattern

2.1 Overview of Measurement and Refinement

The Profex export `exfile1-refined.xy` contains the measured (**observed**) XRD profile alongside the profile calculated from the refined structural model. The experimental scan covers $2\theta = 4.0075°$ to $59.9725°$, with measurement points approximately $0.015°$ apart.

Rietveld refinement was conducted using BGMN within Profex. The refined structural model used a single hydroxyapatite phase operating in hexagonal space group $P6_3/m$ with lattice parameters:

$$a = b = 0.945600 \text{ nm} = 9.45600 \text{ Å}, \quad c = 0.688477 \text{ nm} = 6.88477 \text{ Å} \tag{2}$$

In the refinement plot (Figure 5), the observed profile represents experimental measurement data, the calculated curve represents the fitted structural model, and the bottom difference curve tracks fit residuals.

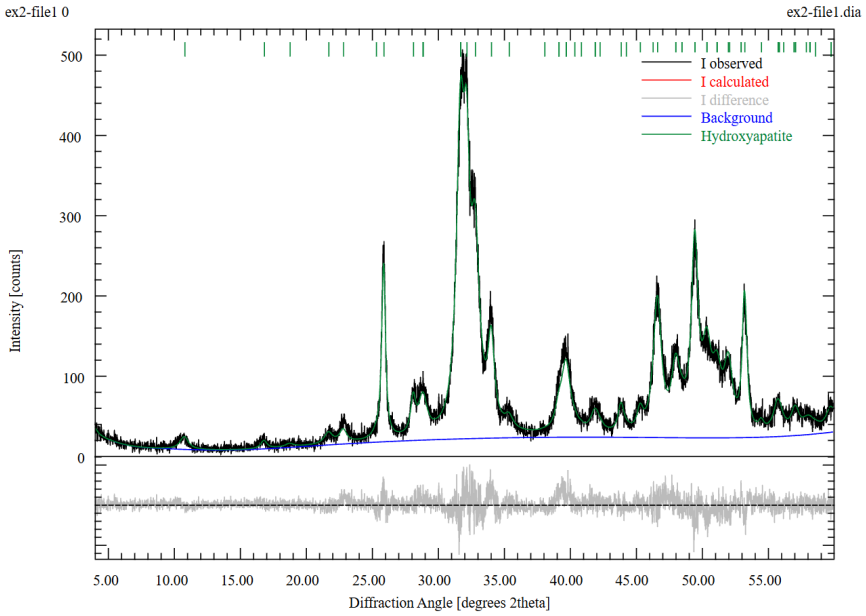


Figure 5: Profex Rietveld refinement plot showing observed and calculated profiles, background curve, difference curve, and reflection markers.

2.2 Main Diffraction Peaks

Table 2 compares the main diffraction peaks in the experimental (observed) and calculated profiles. Intensities are approximate peak heights in counts, rounded to reflect overlapping peaks and small fluctuations between measurement points.

The dominant feature in both profiles is the intense triplet cluster between $31.9°$ and $32.9°$ 2θ. Secondary reflections at $25.9°$, $46.6°$, and $49.5°$ provide crucial fingerprint confirmation.

2.3 Peak Positions and What They Tell Us

X-ray diffraction peak positions are related to the spacing between crystal planes. Agreement between the observed and calculated peak positions indicates that the refined structural model describes the sample's unit-cell dimensions and symmetry reasonably well.

The strongest observed peaks occur near $31.90°$ and $32.13°$ in 2θ, while the calculated peaks occur near $31.77°$ and $32.11°$. Other peaks appear near $25.9°$, $34.0°$, $39.6°$, $46.6°$, $49.5°$, and $53.2°$. The

Table 2: Comparison of main observed and calculated diffraction peaks for the Hydroxyapatite sample.

Main Peak or Region	**Observed Profile (Position & Intensity)**	**Calculated Profile (Position & Intensity)**	**Relative Description**
Peak near $25.9°$	$25.89°$, about 250 counts	$25.89°$, about 230 counts	Prominent peak
Strong peak near $31.9°$	$31.90°$, about 480 counts	$31.77°$, about 470 counts	Main cluster apex
Strong peak near $32.1°$	$32.13°$, about 480 counts	$32.11°$, about 460 counts	Strong overlapping peak
Overlapping group ($32.6°$–$32.8°$)	Peaks near $32.59°$ & $32.79°$ ($340, 330$ counts)	Main peak near $32.74°$, about 320 counts	Coalesced envelope
Peak near $34.0°$	$34.03°$, about 180 counts	$34.01°$, about 160 counts	Moderate intensity
Peak near $39.6°$	$39.64°$, about 140 counts	$39.68°$, about 120 counts	Weaker reflection
Peak near $46.6°$	$46.63°$, about 210 counts	$46.63°$, about 200 counts	Prominent peak
Peak near $49.5°$	$49.46°$, about 260 counts	$49.46°$, about 280 counts	Prominent peak
Peak near $53.2°$	$53.22°$, about 190 counts	$53.22°$, about 200 counts	Moderate intensity

overall agreement in peak positions supports the hydroxyapatite model, although small differences remain, especially in the overlapping group near 32°.

2.4 Peak Intensities and What They Tell Us

Peak intensity depends on the crystal structure and on sample and measurement effects, including overlapping reflections, preferred orientation of crystallites, crystallite size, and instrument conditions.

The strongest observed intensity is about 480 counts in the peak group near 32°. Other prominent observed peaks are near 25.9° (about 250 counts), 46.6° (about 210 counts), and 49.5° (about 260 counts). The calculated profile gives about 470 counts near 31.8° and 460 counts near 32.1°, broadly matching the observed intensities. Peak heights in the 32° region are approximate because the peaks overlap. Observed photon counts represent raw measurement values and should not be confused with normalized theoretical relative intensities.

2.5 Agreement Between Experimental and Calculated Patterns

The calculated pattern follows the main experimental peak positions and overall intensity distribution, particularly the strong peak group near 32°. This suggests that the refined hydroxyapatite model reproduces the principal features of the measured scan.

However, the fit is not exact: the difference curve shows residual discrepancies. Quantitative fit metrics reported in the refinement listing (`.lst`) include:

- Weighted profile factor: $R_{\text{wp}} = 13.72\%$
- Profile factor: $R_{\text{p}} = 10.06\%$
- Expected factor: $R_{\text{exp}} = 12.34\%$
- Bragg factor: $R = 9.46\%$
- Goodness of fit (χ^2): 1.2362

Since R_{wp} is greater than R_{exp}, some mismatch remains between the calculated and experimental patterns. These statistics describe the overall fit quality but do not identify the specific causes of the residuals.

2.6 CIF File Definition and Utility in Structural Analysis

The CIF (Crystallographic Information File) stores a crystal structure in a standardized format that crystallography programs can read. The CIF makes the refined structure portable and reusable. You can open it in VESTA to view the unit cell and atom arrangement, or use it in other crystallography programs for further structural analysis. Programs can also calculate a theoretical diffraction pattern from the structure stored in the CIF.

Summary of key crystallographic information stored in the refined CIF:

- **Symmetry:** Hexagonal, space group $P6_3/m$ (No. 176).
- **Cell Constants:** $a = b = 9.45600$ Å, $c = 6.88477$ Å, $\alpha = \beta = 90^\circ$, $\gamma = 120^\circ$.
- **Atomic Sites:** Calcium (Ca), Phosphorus (P), Oxygen (O), and Hydrogen (H).
- **Portability & Simulation:** Allows 3D structural rendering in VESTA and theoretical diffraction calculations across external crystallographic platforms.

3 Understanding XRD Patterns: Experimental Data vs. Theoretical Simulations

This section compares experimental XRD patterns fitted in Profex against theoretical patterns calculated by VESTA from the refined CIF.

3.1 Key Concepts in X-Ray Diffraction

VESTA calculates diffraction assuming an ideal, infinite, defect-free crystal:

- **Crystallite Size and Peak Broadening:** Real samples consist of finite nanocrystalline domains. Small crystallites broaden diffraction peaks according to the Scherrer equation. VESTA assumes infinite crystallite size, generating sharp, delta-like peaks.
- **Anisotropic Growth:** Crystals growing as rods or needles exhibit direction-dependent domain sizes, causing anisotropic broadening across different (hkl) planes.
- **Microstrain:** Lattice microstrain and defect stresses broaden profiles further.
- **Instrumental Broadening & Noise:** Real diffractometers introduce optical aberration, slit adjustments, and background scattering.

3.2 Overview of the Hydroxyapatite Sample

Refinement parameters used for pattern simulation:

- **Space Group:** Hexagonal $P6_3/m$.
- **Lattice Constants:** $a = b = 9.45600$ Å, $c = 6.88477$ Å.
- **Anisotropic Domain Size:**
  - Along c-axis $[001] = 26.14$ nm
  - Along a-axis $[100] = 10.48$ nm
- **Morphology:** Rod-like nanocrystals elongated along the $[001]$ direction (more than twice as long as wide).

3.3 Agreement in Peak Positions (2θ)

Superimposing VESTA and Profex patterns reveals exact alignment in peak positions (2θ). Because VESTA imports unit-cell parameters determined during refinement, calculated interplanar d-spacings match Bragg positions identically.

3.4 Comparison of Major Diffraction Features

Table 3 presents a detailed reflection-by-reflection comparison.

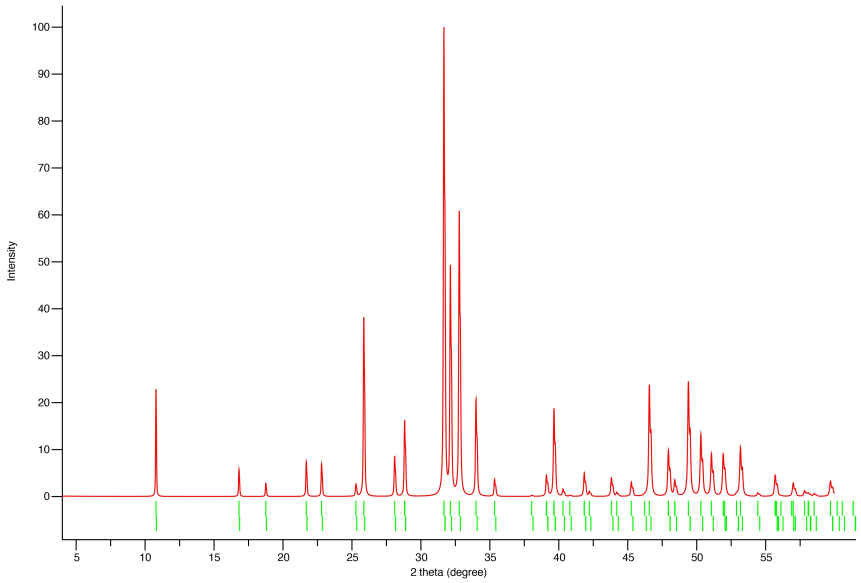


Figure 6: Powder diffraction pattern calculated from the refined CIF using VESTA (relative intensities).

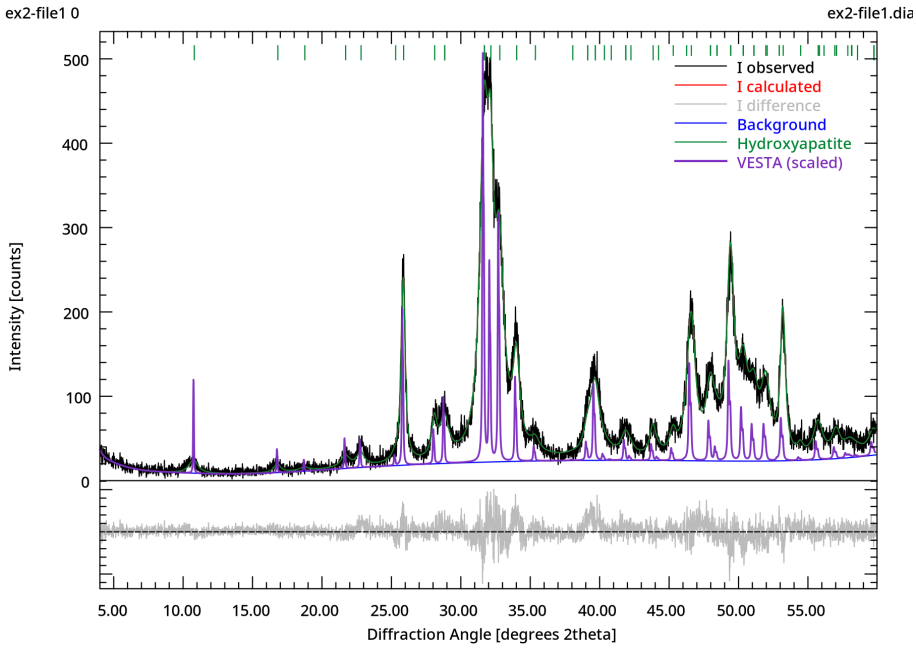


Figure 7: Overlay of experimental Profex display and VESTA-calculated curve (scaled for visual comparison).

Table 3: Detailed comparison between theoretical simulation (VESTA) and experimental refinement (Profex).

Miller Indices (hkl)	**Position** (2θ)	**VESTA (Simulation)**	**Profex (Experimental)**	**Notes**
(100)	10.80°	Sharp, very prominent	Extremely broad, weak hump	Major discrepancy due to Automatic Divergence Slit geometry.
(002)	25.88°	Sharp peak	Relatively sharp, tall peak	Larger domain size along $[001]$ (26.14 nm) reduces broadening.
(211)	31.74°	Distinct spike (100% max)	Forms apex of broad cluster	Primary reflection; overlaps neighboring peaks in real data.
(112)	32.18°	Distinct peak	Merged into side of (211)	Scherrer broadening coalesces adjacent reflections.
(300)	32.88°	Distinct peak	Broad right shoulder	Smaller domain along $[100]$ (10.48 nm) causes heavy broadening.
(004)	53.15°	Weak sharp peak	Broad, overlapped cluster	Overlaps adjacent planes at higher angles.
High-angle $(hk0)$	$\sim 48^\circ - 52^\circ$	Separate sharp peaks	Broadened, overlapping profiles	Multiple $(hk0)$ planes exhibit severe broadening.

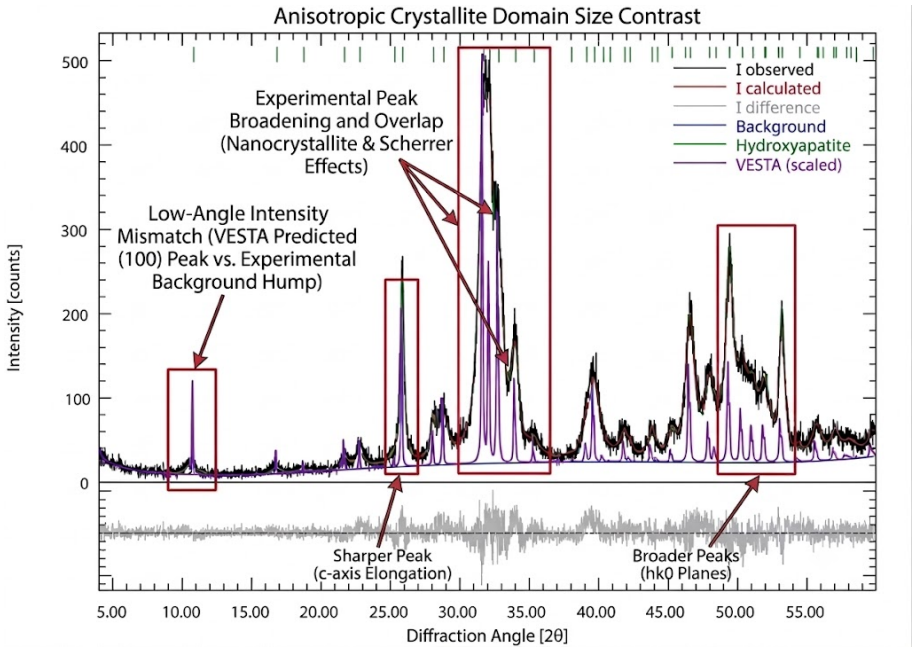


Figure 8: Annotated comparison highlighting profile discrepancies between VESTA and Profex patterns.

3.5 General Differences: Theory vs. Reality

1. **Peak Overlap & Coalescence:** Finite crystallite size broadens reflections, turning discrete theoretical peaks (e.g., $31.74°$, $32.18°$, $32.88°$) into a continuous envelope.
2. **Anisotropic Line Shapes:** Reflections along crystallite length ((002) at $25.88°$) remain sharp, while transverse reflections ((300) at $32.88°$) broaden into wide humps.
3. **Background Baseline:** Real patterns sit on air-scattering background noise, whereas theoretical curves start from zero baseline.

3.6 Major Discrepancy at $2\theta \approx 10.8°$

At low angle ($2\theta \approx 10.8°$, corresponding to (100) plane diffraction), VESTA predicts a sharp peak reaching $\sim 25\%$ maximum intensity. In experimental data, this reflection appears as a faint background bump.

3.6.1 The Limitation of Scherrer Broadening

Scherrer broadening from the 10.48 nm transverse width flattens peak height, but integrated peak area remains lower than predicted by simple geometry.

3.6.2 Primary Instrument Effect: Divergence Slits

The primary cause of intensity loss is the goniometer's **Automatic Divergence Slit (ADS)**:

- VESTA assumes fixed slit geometry, overestimating low-angle irradiated area.
- The experimental diffractometer closed its ADS at shallow angles to keep beam footprint contained within the 10 mm sample boundaries. Restricting beam area physically reduced photon counts at low angles.

- Profex includes ADS optic corrections, accurately matching the suppressed experimental peak.

4 Structural Analysis Report: Hydroxyapatite

4.1 Overview of the Crystal Structure

Hydroxyapatite crystallizes in a hexagonal lattice:

- **Crystal System:** Hexagonal
- **Space Group:** $P6_3/m$ (Space Group No. 176)

4.2 Lattice Parameters and Cell Volume

Unit cell dimensions derived from refinement:

- **Cell Edges:** $a = b = 9.45600$ Å, $c = 6.88477$ Å
- **Interaxial Angles:** $\alpha = \beta = 90°$, $\gamma = 120°$
- **Unit-Cell Volume:**

$$V = a^2 c \sin(120°) = (9.45600)^2 \times (6.88477) \times \frac{\sqrt{3}}{2} = 533.13233 \text{ Å}^3 \tag{3}$$

4.3 Atomic Arrangement and Structural Features

Rendering the CIF in VESTA illustrates atom locations and bonding polyhedra.

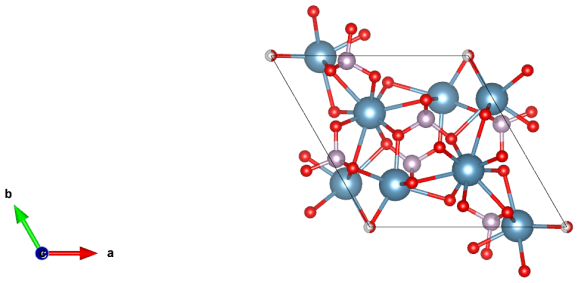


Figure 9: VESTA rendering of hydroxyapatite structural model showing unit cell, atomic species, and crystallographic axes.

Key structural findings:

- **Framework Stability:** Calcium (Ca1, Ca2), Phosphorus (P3), and Oxygen (O4, O5, O6) sites show full occupancy (1.0000).
- **Phosphate Tetrahedra:** P atoms form rigid PO_4 tetrahedra serving as structural pillars.
- **Hydroxyl Channels & Disorder:** Hydroxyl groups (OH^-) align along c-axis channels. Mirror symmetry across space group $P6_3/m$ requires hydroxyl oxygen (O7) and hydrogen (H8) to exhibit fractional occupancy (0.5000), representing orientational disorder.

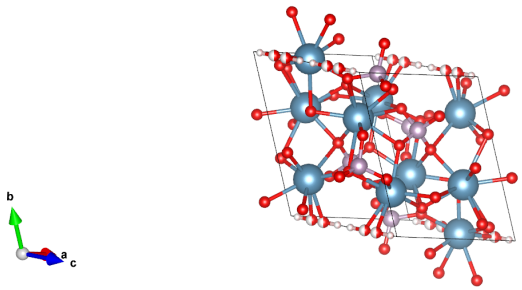


Figure 10: Alternative view of the hydroxyapatite unit cell showing atom arrangements along different crystallographic projections.

Table 4: Color key and structural descriptions for atomic species rendered in VESTA.

Colour in VESTA	**Atom Species**	**Structural Description**
Large blue spheres	Calcium (Ca)	Located between phosphate tetrahedra, coordinated by oxygen atoms.
Pale purple/pink spheres	Phosphorus (P)	Tetrahedrally coordinated by four oxygen atoms (PO_4^{3-}).
Red spheres	Oxygen (O)	Form vertices of phosphate tetrahedra and hydroxyl channels.
Half-filled white spheres	Hydrogen (H)	Located on channel axes, exhibiting partial site occupancy (0.5000).

Table 5: Comparison of refined unit-cell parameters with published literature standards.

Reference Source	**a-axis (Å)**	**c-axis (Å)**	**Cell Volume (Å^3)**
Refined Model (This Work)	**9.4560**	**6.8848**	**533.13**
Kay, Young & Posner (1964) [1]	9.4320	6.8810	530.14
Sudarsanan & Young (1969) [2]	9.4240	6.8790	528.80
ICSD Standard #26204 [8]	9.4180	6.8840	528.53
RRUFF Reference R060180 [7]	9.4193	6.8792	528.58

4.4 Comparison with Published Reference Values

Table 5 compares refined cell parameters against benchmark literature.

4.5 Discussion and Structural Conclusions

The refined c-axis parameter (6.88477 Å) matches historical literature values within 0.05%. The a-axis (9.45600 Å) exhibits a slight lattice expansion of ~ 0.024 Å (~ 0.25%), increasing calculated volume to 533.13 Å^3. Such minor expansions are standard, arising from measurement temperature (295 K), model background parameters, or trace lattice moisture/substitutions.

5 Overall Summary and Conclusions

Experimental XRD data from `ex2-file1.xrdml` were analyzed using Profex, and the hydroxyapatite crystal structure was refined using the Rietveld method within BGMN. Primary reflections cluster between 31° and 34° 2θ, supplemented by reflections at 25.9°, 46.7°, and 49.5°.

Rietveld refinement metrics ($R_{\text{wp}} = 13.72$%, $R_{\text{p}} = 10.06$%, $R_{\text{exp}} = 12.34$%, $\chi^2 = 1.2362$) confirm robust fit quality. The refined structural model operates in hexagonal space group $P6_3/m$ with cell parameters $a = b = 9.45600$ Å, $c = 6.88477$ Å, and volume $V = 533.13$ Å^3.

Simulations in VESTA replicate fundamental peak positions while highlighting real-world physical discrepancies:

1. Nanocrystalline Scherrer broadening causes peak overlap in experimental data.
2. Anisotropic grain dimensions (26.14 nm along $[001]$ vs. 10.48 nm along $[100]$) produce directional line-shape differences.
3. Instrument optics (Automatic Divergence Slits) account for low-angle intensity suppression at $2\theta \approx 10.8^\circ$.

In summary, combining raw XRD data parsing, Rietveld refinement, CIF export, and VESTA 3D visualization provides a comprehensive structural evaluation of hydroxyapatite.

6 Individual Contributions

- **Saraangan:** Managed the software workflow using VESTA and Profex, performed the Rietveld refinement and crystal structure generation, and coordinated the team's progress (Overall and Section 5).
- **Tejas:** Validated the software-generated results and conducted the comparison of the experimental and calculated XRD patterns obtained from Profex and VESTA (Section 3).
- **Snigdha:** Analysed the experimental hydroxyapatite XRD data, described the data file and its characteristics, and discussed the crystal structure of hydroxyapatite (Sections 1 and 4).
- **Ansika:** Analysed the refined XRD pattern and explained the definition, purpose, and utility of the Crystallographic Information File (CIF) in structural analysis (Section 2).
- **Samarth:** Assisted with the final review of the report by checking the consistency of the structural parameters, figure captions, and references.

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